E-Commerce Product Gallery – Project Documentation

Project Information

Project Name: E-Commerce Product Gallery
Developer: Gurusha Dholwani
Domain: Frontend Web Development
Duration: 4 Weeks

Project Objective

The objective of this project is to develop a responsive E-Commerce Product Gallery that dynamically fetches and displays products using an external API. The application allows users to browse products, search items, filter products by category, and view detailed information through an interactive modal interface.

The project demonstrates modern frontend development concepts including API integration, dynamic rendering, responsive design, accessibility improvements, and performance optimization.

Problem Statement

Modern e-commerce applications require dynamic content presentation and responsive user interfaces. Static websites cannot efficiently manage large product collections or provide interactive browsing experiences.

This project aims to solve these challenges by integrating a product API and implementing dynamic search, filtering, and product viewing functionality.

Project Scope

Included Features

- Product Listing
- Dynamic Product Rendering
- Fake Store API Integration
- Product Search
- Category Filtering
- Product Details Modal
- Loading States
- Error Handling
- Empty States
- Product Statistics

- Responsive Design
- Accessibility Enhancements

Excluded Features

- Shopping Cart
- Payment Gateway
- Authentication System
- Backend Database
- Order Management

Technologies Used

- HTML5
- CSS3
- Bootstrap 5
- JavaScript (ES6+)
- Fetch API
- Fake Store API
- Git
- GitHub

Folder Structure

```
E-Commerce-Product-Gallery/

 ├── index.html

 ├── css/
 │   └── style.css

 ├── js/
 │   └── script.js

 ├── assets/
 │   ├── HOME.png
 │   ├── PRODUCTS.png
 │   ├── CATEGORY.png
 │   ├── VIEW DETAIL.png
 │   └── FOOTER.png

 ├── documentation/
```

```
├── screenshots/
├── README.md
└── .gitignore
```

Application Workflow

1. User opens the application.
2. Product data is fetched from Fake Store API.
3. Products are rendered dynamically.
4. Statistics are generated automatically.
5. Users search products by keywords.
6. Users filter products by category.
7. Users view product details through the modal.
8. Loading, error, and empty states provide feedback.
9. Responsive layouts adapt to different screen sizes.

Features Implemented

Product Gallery

- Dynamic product cards
- Product images
- Product prices
- Product ratings
- Category labels

Search Functionality

- Real-time product search
- Case-insensitive matching
- Search by title and category

Category Filtering

- Electronics
- Jewelery
- Men's Clothing
- Women's Clothing

- All Products

Product Details Modal

- Product image
- Product title
- Product description
- Product category
- Product rating
- Product price

User Experience Features

- Loading spinner
- Skeleton loading cards
- Empty state messaging
- Error handling
- Smooth scrolling
- Active navigation highlighting
- Back-to-top button

Accessibility Features

- Semantic HTML structure
- Skip navigation link
- Focus-visible styles
- Keyboard navigation support
- ARIA labels and live regions
- Reduced motion support

Screenshots

Home Page

assets/HOME.png

Product Gallery

assets/PRODUCTS.png

Category Filters

assets/CATEGORY.png

Product Details Modal

assets/VIEW DETAIL.png

Footer

assets/FOOTER.png

Performance Optimizations

- Lazy-loaded images
- Async image decoding
- Efficient DOM rendering
- DocumentFragment usage
- Reduced reflows and repaints
- Skeleton loading placeholders
- Passive event listeners

Learning Outcomes

Through this project, the following skills were strengthened:

- API Integration
- Fetch API
- Async/Await
- DOM Manipulation
- Event Handling
- Search and Filtering Logic
- Responsive Design
- Bootstrap Components
- Accessibility Best Practices
- Performance Optimization
- Git & GitHub Workflow
- Project Documentation

Future Enhancements

- Shopping Cart

- Wishlist Functionality
- Product Sorting
- Pagination
- User Authentication
- Product Reviews
- Dark Mode
- Backend Integration
- Payment Gateway

Conclusion

The E-Commerce Product Gallery successfully demonstrates modern frontend development concepts through a responsive, accessible, and user-friendly interface.

The project integrates real-time product data from an external API, implements search and category filtering, provides detailed product views through a modal, and incorporates accessibility and performance best practices.

This project serves as a strong portfolio project showcasing practical frontend development, API integration, JavaScript programming, responsive design, and professional GitHub workflow.